



# An ISO-Informed Methodological Audit of Life Cycle Assessment Practices in Recycled Aggregate Concrete

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## Abstract

Concrete waste recycling is increasingly recognized as a strategy for reducing the environmental burdens associated with end-of-life construction materials, landfilling, and virgin material extraction. Within the transition to a circular economy, recycled concrete aggregate has gained growing attention as a secondary aggregate resource for concrete production. However, before recycled aggregate concrete (RAC) can be widely promoted as an environmentally preferable alternative, its environmental performance must be evaluated through rigorous, transparent, and standardized life cycle assessments (LCAs). This paper presents a PRISMA-guided systematic review of 37 RAC LCA studies and conducts an ISO-informed methodological audit across the four phases of the ISO 14040/44 framework. The review identifies recurring limitations: 82% of system boundary instances were limited to cradle-to-gate, constraining full life cycle evaluation; 50% of functional unit instances adopted a volumetric unit of 1 m<sup>3</sup> of concrete, which may not fully capture functional equivalence; and 65% of studies did not report how allocation and multifunctionality were handled, limiting transparency and reproducibility. Additional gaps were observed in inventory data representativeness, impact assessment method reporting and justification, interpretation practices, and data quality analysis. By linking identified limitations to ISO-aligned recommendations, this review provides practical guidance for improving the robustness, transparency, comparability, and credibility of future RAC LCA studies and circularity claims.

**Keywords** Life cycle assessment · ISO 14040/44 · Recycled aggregate concrete · Construction and demolition waste · Circular economy · Methodological audit

## Introduction

Construction and demolition waste (CDW) remains one of the largest waste streams, accounting for a substantial share of total solid waste in many regions [1, 2]. CDW comprises a heterogeneous mix of inert and non-inert materials, as well as hazardous and non-hazardous streams, typically including concrete, clay bricks, mortar, wood, plastics,

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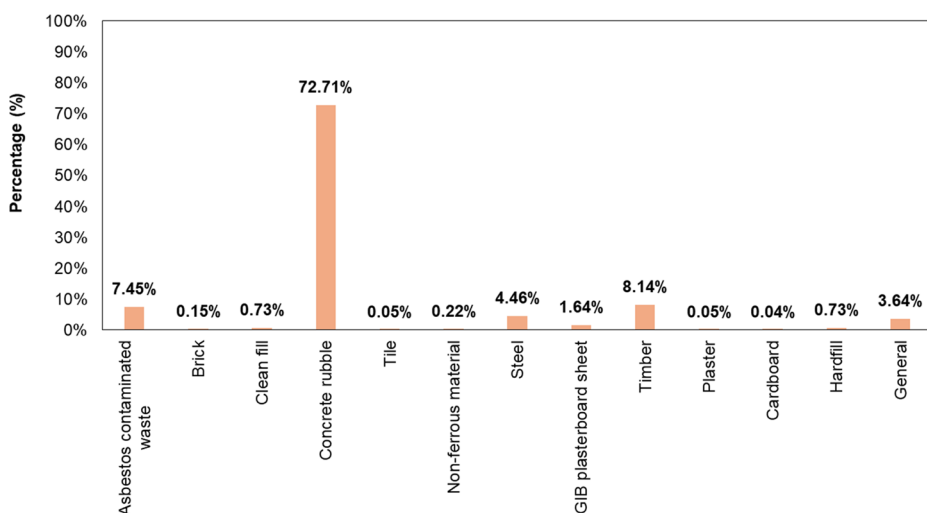
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and steel [3]. CDW generation varies across regions, with China, the United States, India, and the European Union as the dominant contributors [4]. With population growth, rapid urbanization, and continued infrastructure expansion, CDW management is expected to remain a critical challenge in the coming decades [5].

Recent policy frameworks increasingly promote the reduction, reuse, and recycling of CDW to reduce the extraction of virgin materials and lower the environmental impacts associated with primary production and transportation [5]. Given the high global demand for concrete [6] and the high repurposing potential of CDW [7], incorporating recycled CDW into concrete offers a promising sustainable pathway to support circular construction. A key advantage of CDW recycling is reduced extraction of natural aggregate (NA), thereby limiting new mining [8, 9]. CDW recycling also contributes to waste diversion and can reduce the environmental burdens associated with landfilling [10, 11].

The majority of CDW consists of concrete waste (CW), typically exceeding 70% (Fig. 1). Consequently, CDW recycling primarily involves CW recycling. In this context, research and practical applications have predominantly focused on recycled coarse aggregate (RCA) as a substitute for natural coarse aggregate (NCA) [9, 12]. Existing technical standards predominantly provide guidance for RCA use in concrete, while providing limited provisions for recycled fine aggregate (RFA) or recycled concrete powder (RCP) [13]. Therefore, recycled aggregate concrete (RAC) incorporating RCA provides a practical and comparatively mature reference case for refining and harmonizing methodological practices in life cycle assessment (LCA) studies of recycled concrete products.

LCA is widely used as a systematic approach for quantifying the environmental impacts of a product system [15]. In construction, LCA is increasingly used to inform early-stage design decisions and to evaluate end-of-life (EOL) strategies, including



**Fig. 1** Overall CDW waste composition, based on data reported by Peiris et al. [14]

recycling, reuse, and disposal [5, 16, 17]. Applications of LCA to RAC have a relatively long research history, with studies demonstrating that the environmental performance of RCA substitution is highly sensitive to LCA modeling assumptions. For example, early studies have shown that system boundary definition, transport distance, allocation approach, avoided-burden modeling, functional equivalence, recycling system design, and recycling-loop representation can substantially influence LCA outcomes [18–21]. These early contributions indicate that methodological variability is not a recent issue, but a long-standing challenge in RAC LCA studies.

Despite this long research history, RAC LCA outcomes remain insufficiently robust to support confident investment in high-quality RCA production for upcycling CW for application in new concrete rather than downcycling into road base or fill [9]. While RAC mix design and mechanical performance have been extensively investigated, environmental assessment has received comparatively less attention, and studies that apply LCA often consider only a limited number of mix designs [22].

Although LCA has been standardized by the International Organization for Standardization (ISO), ISO 14040:2006 [23] and ISO 14044:2006 [24] provide a general LCA framework rather than prescriptive rules for key methodological choices, including functional unit (FU) definition, system boundary selection, life-cycle stage inclusion or exclusion, allocation approaches, and interpretation [25, 26]. Consequently, LCA practitioners often apply diverse methodological assumptions, creating ambiguity regarding the comparative environmental performance of RAC systems [25]. This issue is particularly important in the context of the circular economy (CE), where the distinction between closed-loop and open-loop recycling can substantially influence the interpretation of LCA results.

Reflecting these challenges, previous review studies have examined LCA practices in CDW management, CW recycling, and RAC systems, revealing significant methodological inconsistencies. Zhang et al. [27] identified key sources of variability in RAC LCA studies, including mix design method, FU definition, inventory allocation, CO<sub>2</sub> uptake through carbonation, and transport distance. Xing et al. [28] similarly highlighted that RAC LCA outcomes are strongly influenced by variations in the mix design and LCA modeling mechanisms. Marinković et al. [26] further emphasized the need to connect the sustainability assessment of RAC systems with LCA methodologies, standards, and design practice. Broader CDW and CW recycling reviews have also reported inconsistencies in FU definition, system boundary selection, impact category selection, life cycle impact assessment (LCIA) methods, CE integration, allocation modeling, inventory data quality, and recycled material quality considerations [5, 25, 29, 30].

Existing reviews have primarily focused on identifying variations in methodological choices and comparing their influence on LCA. Limited attention has been given to critically evaluating whether the adopted methodological choices in RAC LCA studies are transparently reported, adequately justified, consistently applied, and reproducible in accordance with the widely acknowledged ISO 14040 [23] and ISO 14044 [24] principles. This represents an important gap as studies may claim ISO compliance yet provide insufficient methodological detail to support reproducibility, comparability, and decision-relevant interpretation.

To address this gap, the present study conducts an ISO-informed methodological audit of RAC LCA studies, evaluating key methodological components including FUs, system boundaries, allocation and multifunctionality handling, inventory development, LCIA methodologies, software tool use, interpretation practices, and data quality assessment. The audit identifies persistent inconsistencies in the operationalization of ISO 14040/44 principles across RAC LCA studies. Building on these findings, the study develops ISO-aligned and RAC-specific methodological guidance to support more transparent, consistent, reproducible, and decision-relevant LCA outcomes for circular concrete systems.

## Methods

This research employs a systematic review approach to critically examine contemporary LCA practices for RAC. The review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to guide screening, selection, and reporting, as it is widely used in systematic reviews to ensure transparent and systematic evidence reporting [31–33].

### Literature Search Strategy

The literature search was primarily conducted using the Scopus database, which is one of the world's largest curated databases of high-quality peer-reviewed research [34]. A comprehensive search query was developed for the literature search process using key terms including 'construction and demolition waste', 'life cycle assessment', and 'recycl\*', together with related keywords (Supplementary Information S1).

### Study Selection Criteria and Screening Process

The review included peer-reviewed English-language journal articles published between January 2015 and May 2025 to capture contemporary LCA practice and recent methodological advances. The initial Scopus search yielded 302 records, and three additional studies were identified through backward and forward snowballing. The title and keyword screening stage focused on retaining studies that explicitly addressed CDW management and recycled concrete with an LCA focus, resulting in 224 records. The abstract screening stage focused on retaining studies that appeared to assess RCA integration in concrete from an LCA perspective, yielding 140 records. During full-text screening, 103 records were excluded where detailed review showed that they did not focus on recycled concrete LCA, did not incorporate RCA as a recycled input material, or lacked core LCA methodological information, including the FU, system boundary, and LCIA, to enable meaningful extraction, classification, and comparison within the audit framework. The full screening and eligibility criteria are provided in Supplementary Table S1. Following systematic screening and eligibility assessment, 37 studies were retained for in-depth review and are listed in Supplementary Table S2. The full screening process is summarized in the PRISMA flow diagram (Fig. 2).

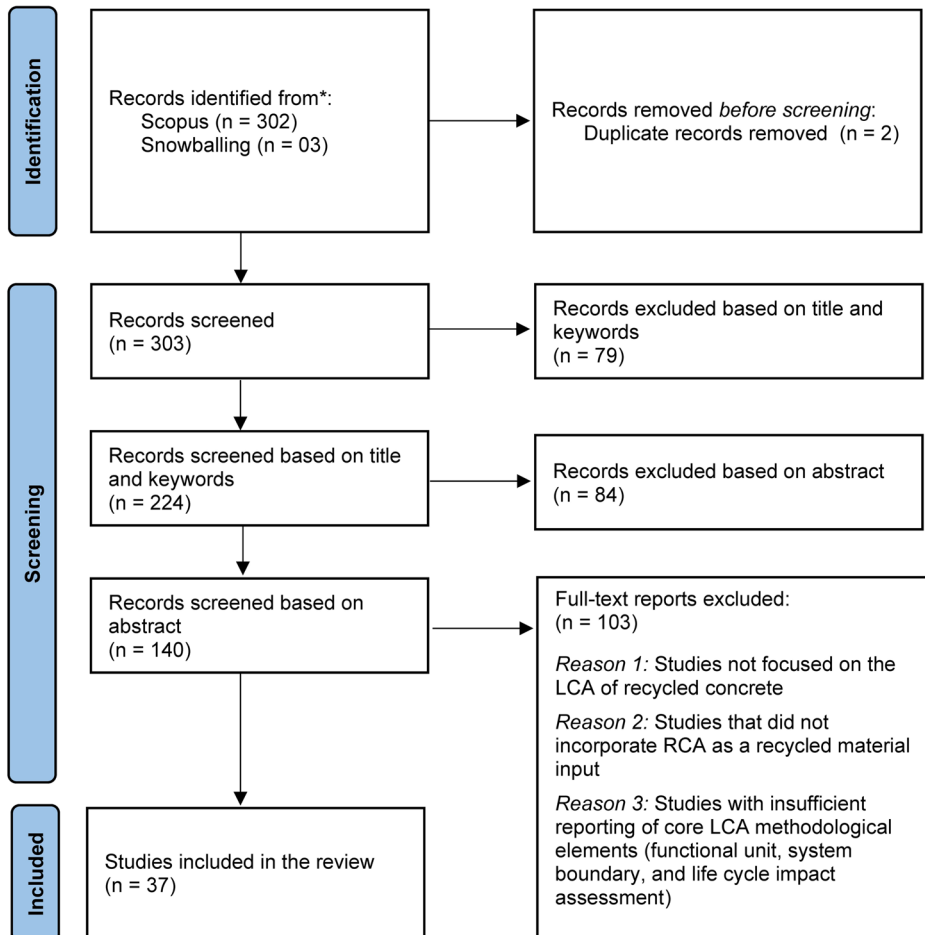


Fig. 2 PRISMA flow diagram for study identification, screening, eligibility assessment, and inclusion

## Data Extraction, Organization, Counting, and Reporting Conventions

### Data Extraction and Organization

The final dataset was systematically extracted to capture descriptive and methodological information from the reviewed studies. Key extracted variables included study descriptors, such as keywords, publication year, and geographic region, as well as LCA methodological elements, including FUs, system boundaries, allocation and multifunctionality handling, life cycle inventory (LCI) data sources, LCIA methods, impact categories, software tools, interpretation practices, and data quality analysis. This structured extraction process formed the evidence base for the descriptive overview, quantitative summary, and qualitative methodological synthesis.

## Counting and Reporting Convention (Study-Level vs. Instance-Level)

Because some methodological elements were reported more than once within a single study (e.g., multiple FUs), both study-level and instance-level counts were used in the quantitative synthesis. Accordingly, percentages were calculated using the appropriate denominator, and the terminology (“studies” vs. “instances”) is used consistently throughout the paper. For study-level reporting, the default denominator was  $N=37$ . For instance-level reporting, the denominator varies with the total number of reported instances for the relevant element. These totals are clearly indicated in the relevant text, tables, and figures.

## Descriptive Overview and ISO-Informed Methodological Audit

Following the systematic identification, screening, and selection of relevant RAC LCA studies, the analysis combined a descriptive overview of the reviewed studies with an ISO-informed methodological audit. The descriptive overview briefly summarized publication years, keyword patterns, and geographic distribution to contextualize the reviewed studies and provide a general overview of research patterns.

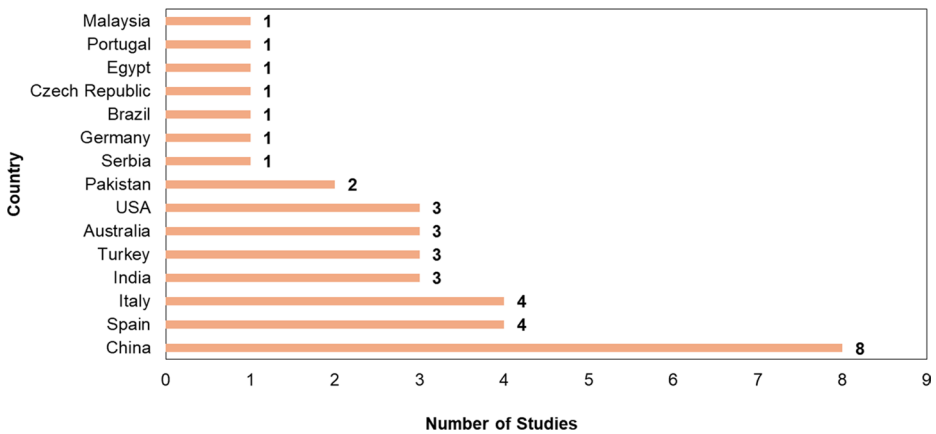
The ISO-informed methodological audit examined RAC LCA methodological practices, structured around the four phases defined in ISO 14040 and ISO 14044: goal and scope definition, life cycle inventory analysis, life cycle impact assessment, and life cycle interpretation [23, 24]. The audit focused on whether LCA methodological choices were explicitly reported, adequately justified, consistently applied, and sufficiently transparent to support reproducibility and comparison across RAC LCA studies. This combined approach enabled the review to integrate a transparent systematic review process, a concise contextual overview, and a detailed qualitative synthesis of key methodological practices, reporting patterns, ISO-informed gaps, and ISO-aligned recommendations.

## Results

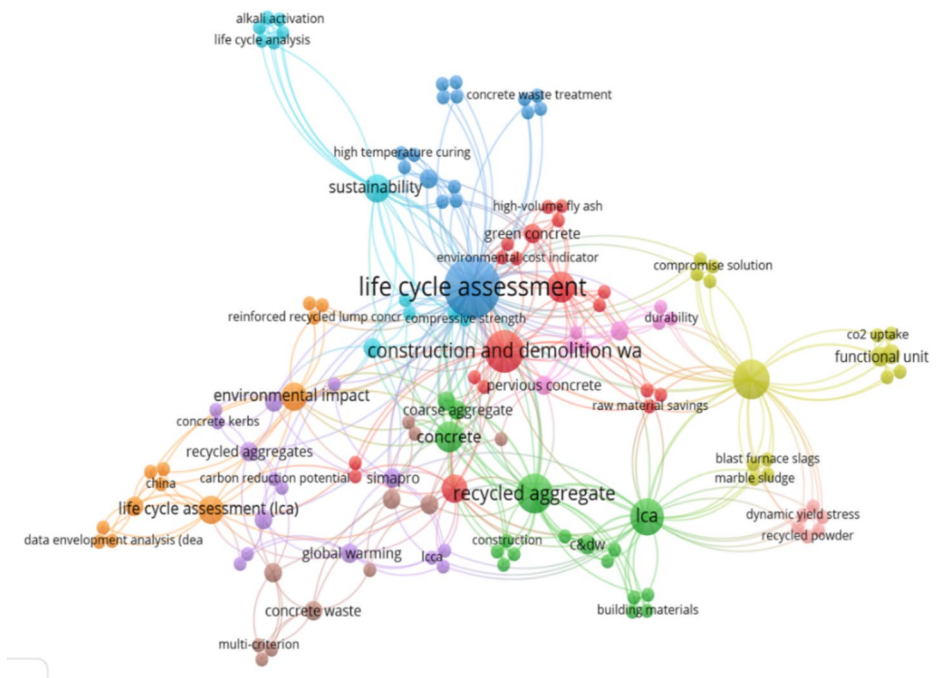
### Descriptive Overview of the Reviewed Studies

Figure 3 illustrates the temporal distribution of RAC LCA studies published over the past decade. Most studies (25) were published between 2021 and 2025, indicating a notable increase in the acceptance and application of LCA as a tool for assessing the environmental sustainability of recycled concrete systems. According to the geographical distribution shown in Fig. 4, China leads with 8 studies, followed by Italy and Spain, each with 4 studies. Other countries, including the USA, India, Australia, and Turkey, contributed 3 studies each, while several additional countries are also represented with fewer studies.

Figure 5 presents the keyword co-occurrence map of the reviewed studies, highlighting the most prominent and interconnected research themes. Keyword co-occurrence analysis is useful for identifying relationships among frequently used terms and for revealing thematic patterns within the literature [35]. Larger nodes indicate higher keyword frequency, links represent co-occurrence relationships, and different colors indicate thematic clusters. The most frequently occurring author keywords were “life cycle assessment”, “construction and

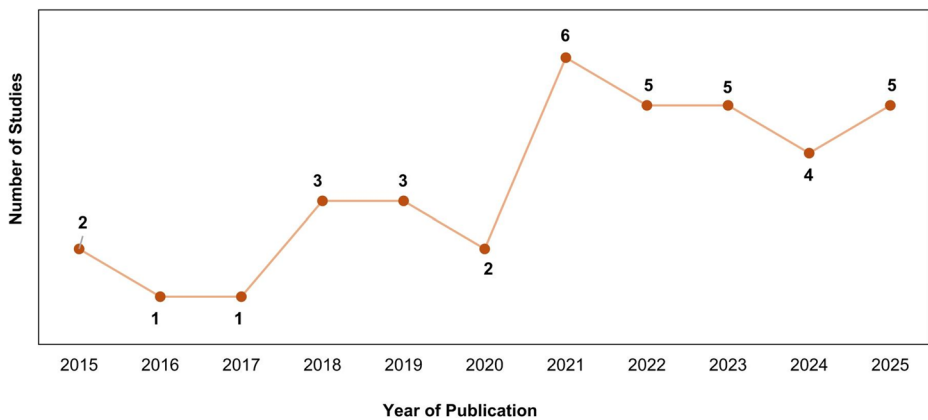


**Fig. 4** Geographical distribution of the reviewed RAC LCA studies (studies,  $N=37$ )



**Fig. 5** Keyword co-occurrence map of the reviewed studies (studies,  $N=37$ ). The map was generated using VOSviewer based on extracted author keywords with a minimum keyword occurrence threshold of two

demolition waste”, “recycled aggregate”, “recycled concrete aggregate”, and “sustainability”, confirming that the reviewed studies are strongly centered on environmental assessment of recycled concrete and CDW recovery pathways.



**Fig. 3** Annual distribution of reviewed RAC LCA studies published between January 2015 and May 2025 (studies,  $N=37$ )

## LCA Methodological Practices

LCA is defined by ISO as the “compilation and evaluation of the inputs, outputs, and the potential environmental impacts of a product system throughout its life cycle” [23]. In accordance with ISO 14040 and ISO 14044, an LCA consists of four phases: (i) goal and scope definition, (ii) life cycle inventory analysis, (iii) life cycle impact assessment, and (iv) life cycle interpretation [23, 24]. To enhance the paper’s thematic coherence and analytical clarity, the findings are structured around these phases. Where relevant, reference is also made to the EN 15804:2012+A2:2019 standard, as it provides specific guidance for LCA of construction products [36].

### Goal and Scope Definition

The goal of an LCA should define the intended application, the rationale for undertaking the LCA, the target audience, and whether the findings will be used in comparative assertions disclosed to the public [23]. The scope of an LCA encompasses a wide range of aspects. This review focuses on four principal elements: system approach, FU, system boundary, and allocation and multifunctionality handling.

**System Approach (Attributional/Consequential)** The two main system approaches used in LCA are attributional LCA (ALCA) and consequential LCA (CLCA) [37]. ALCA assesses a product’s environmental impacts within the current system and is commonly used to compare products or processes under existing conditions. In contrast, CLCA assesses the environmental consequences of a decision or change in the product system (e.g., an increase in RCA demand) [25].

Among the reviewed studies, none explicitly stated the system approach adopted. However, based on the background databases used, the FUs defined, and the reported LCA outcomes, 30 studies can reasonably be classified as ALCA studies. The remaining seven studies [9, 22, 38–42] applied a substitution-based approach using avoided-burden credits to address recycling-related multifunctionality. This is generally interpreted as an implementation of system expansion. Although some authors interpret the use of system expansion as an indicator of CLCA [43], Schaubroeck

et al. [37] argue that the system-expansion approach applies to both ALCA and CLCA. Consequently, these seven studies present some ambiguity in classification. Nonetheless, aside from the use of substitution-based avoided-burden modeling, the background databases, FUs, and reported LCA outcomes of these studies also align well with the ALCA system approach.

**Functional Unit (FU)** ISO defines the FU as “quantified performance of a product system for use as a reference unit” [23]. It provides the basis for quantifying all inputs and outputs associated with the product system [25]. Therefore, the FU must be consistent with the goal and scope definition of the study [23]. In this review, six different FUs have been identified (see Table 1). The volumetric FU (1 m<sup>3</sup> of concrete) was the most frequently applied (21 instances), followed by the strength-based FU (11 instances). The complete list of references corresponding to each FU category is provided in Supplementary Table S3.

**System Boundary** Concrete undergoes several life-cycle phases, including the production, construction, use, EOL, and material recovery. Consequently, system boundary selection must be carefully aligned with the study goal and scope, as it determines which stages are included in the LCA modeling [23]. The most common system boundaries in concrete LCAs are cradle-to-gate, cradle-to-grave, and cradle-to-cradle. Cradle-to-gate covers unit processes from raw material extraction to concrete production. Cradle-to-grave extends the scope to include use and maintenance phases and EOL processes (e.g., demolition, transport, and disposal). Cradle-to-cradle further extends the boundary by including EOL recovery and the reintegration of recovered materials into a subsequent product system, thereby enabling CE-related benefits and burdens to be represented. In this review, 31 instances were limited to cradle-to-gate, while cradle-to-grave (4 instances) and cradle-to-cradle (3 instances) boundaries were explored minimally (see Fig. 6(a)).

However, the sustainability of RAC systems cannot be evaluated solely on a cradle-to-gate basis. Given the role of RCA use as a CE strategy, greater attention is needed to extend system boundaries beyond EOL to incorporate cradle-to-cradle scenarios. In such cases, a key challenge is determining how environmental burdens and potential credits should be shared between the previous concrete life cycle and the subsequent life cycle in which recovered CW is processed into RCA and used to produce RAC. This directly relates to the broader discussion of allocation and multifunctionality handling.

**Table 1** Functional units identified in the reviewed studies (instances,  $N=42$ )

| FU Code         | FU Type                          | No. of Instances | Description                                                                                              |
|-----------------|----------------------------------|------------------|----------------------------------------------------------------------------------------------------------|
| FU <sub>1</sub> | Volumetric FU                    | 21               | 1 m <sup>3</sup> of concrete, irrespective of performance properties                                     |
| FU <sub>2</sub> | Strength-based FU                | 11               | 1 m <sup>3</sup> of concrete with equivalent compressive strength                                        |
| FU <sub>3</sub> | Strength-workability FU          | 3                | 1 m <sup>3</sup> of concrete with equivalent compressive strength and workability                        |
| FU <sub>4</sub> | Strength-durability FU           | 3                | 1 m <sup>3</sup> of concrete with equivalent compressive strength and durability                         |
| FU <sub>5</sub> | Performance-normalized impact FU | 3                | Environmental impact normalized by performance parameters (e.g., impact per MPa of compressive strength) |
| FU <sub>6</sub> | Mass-based FU                    | 1                | 1 ton of concrete, irrespective of performance properties                                                |

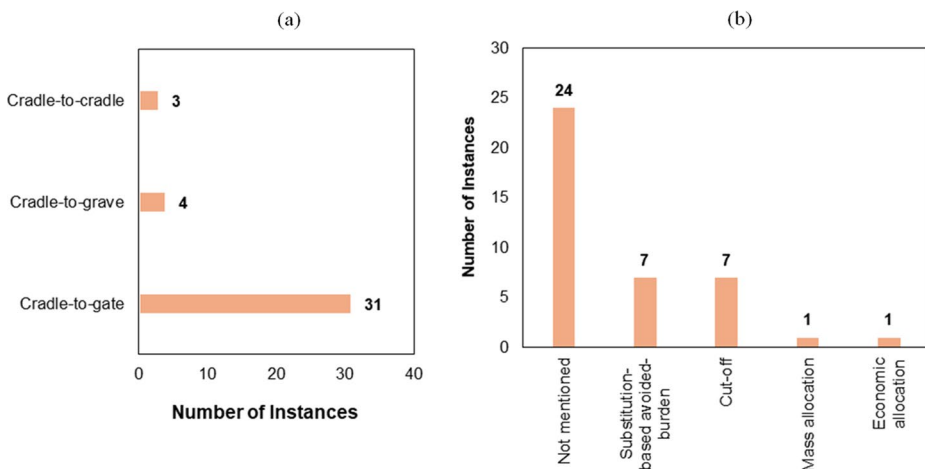
Totals are reported as instances because some studies reported multiple functional units

**Allocation and Multifunctionality Handling** CW recycling commonly generates multiple outputs, with RCA being only one co-product. Where subdivision or system expansion is not feasible, this multifunctionality requires environmental burdens to be allocated among co-products [25]. In addition, CW recycling introduces a system-level multifunctionality, as burdens and potential credits may need to be distributed across consecutive life cycles when recovered CW is processed into RCA and used in new concrete. To handle cross-life-cycle multifunctionality, the primary approaches are cut-off and substitution-based avoided-burden, while approaches based on physical relationships or economic value are less frequently applied. In contrast, allocation within multi-output processes is typically based on measurable physical relationships (e.g., mass, volume, or energy content) or the economic value of co-products [25].

In this review, most studies (24 of 37) did not report how allocation and multifunctionality were handled, creating uncertainty regarding how environmental burdens and potential credits were assigned between co-products and/or across life cycles. Among the studies that specified an approach for cross-life-cycle multifunctionality handling, the substitution-based avoided-burden approach was applied in seven instances, the cut-off approach in seven instances, and economic allocation in one instance. Only one instance addressed multi-output allocation using mass allocation. The distribution of these approaches is shown in Fig. 6(b), while the principles of each approach are summarized in Table 2, and the associated references are listed in Supplementary Table S4.

### Life Cycle Inventory (LCI) Analysis

LCI analysis is the second phase of an LCA [23]. This phase quantifies all inputs and outputs of unit processes within the system boundary, expressed relative to the FU. A robust LCI



**Fig. 6** Distribution of (a) system boundary selections among the reviewed studies (instances,  $N=38$ ) and (b) approaches for allocation and multifunctionality handling in the reviewed studies (instances,  $N=40$ ). Totals are reported as instances because some studies reported multiple system boundaries and/or allocation and multifunctionality-handling approaches

**Table 2** Reported approaches for allocation and cross-life-cycle multifunctionality handling in the reviewed studies (instances,  $N=16$ )

| Category                                     | Approach                          | No. of Instances | Description                                                                                                                                                                                                               |
|----------------------------------------------|-----------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cross-life-cycle multifunctionality handling | Substitution-based avoided-burden | 7                | The system boundary is expanded to account for displaced equivalent virgin material production and, where applicable, avoided waste-management processes, with avoided-burden credits assigned to the assessed RAC system |
|                                              | Cut-off                           | 7                | The previous product system bears the burden of primary material production, while the RAC system bears only the recycling-related impacts. Recycled materials enter the subsequent life cycle burden-free                |
|                                              | Economic allocation               | 1                | Environmental impacts from both the original and recycling stages are shared between the previous and subsequent product systems according to their relative economic value                                               |
| Multi-output allocation                      | Mass allocation                   | 1                | Environmental burdens from multi-output unit processes are apportioned according to the mass fraction of each co-product (e.g., RCA, fines, recovered steel)                                                              |

This table summarises only explicitly reported approaches for allocation and multifunctionality handling. Totals are reported as instances because some studies reported multiple approaches. The broader distribution of these approaches is presented in Fig. 6(b)

depends on both system definition and the quality and completeness of underlying data. The inventory should include datasets representing resource inputs and emission outputs for each unit process [30], and data quality remains a key determinant of reliability.

In the studies reviewed, inventory data were typically derived from multiple types of data sources. Since individual studies often used more than one inventory data source, the reported counts are not mutually exclusive. Overall, 21 studies used primary data, 26 studies used literature-derived data, and 20 studies used the Ecoinvent database. Less frequently used secondary databases included ELCD, CLCD, and GEMIS. The inventory data sources used in each reviewed study are summarized in Supplementary Table S5.

Primary data were mainly obtained through site-specific measurements, interviews, surveys, case studies, monitoring analyses, and direct data collection at concrete production and recycling facilities. Several studies collected context-specific data from recycling facilities, particularly where established databases did not provide sufficient information for RCA production and processing [6, 11]. Some studies also used laboratory-scale recycling experiments and scaled the results to represent industrial equipment or full-scale recycling processes [3, 40]. Secondary data were obtained primarily from proprietary databases, published literature, government and industry reports, and public datasets. Literature-derived inventory data were widely used, while Ecoinvent was the most frequently adopted proprietary database. Some studies also incorporated region-specific electricity grid datasets to improve contextual relevance [44, 45].

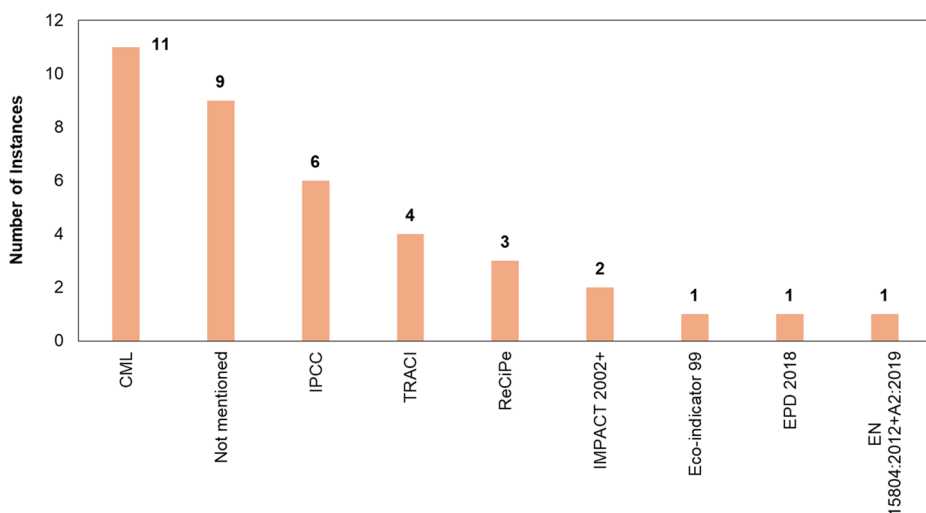
### Life Cycle Impact Assessment (LCIA)

LCIA is the third phase of an LCA [23], and focuses on selecting impact categories, category indicators, and characterization models [25]. In this phase, classification means assigning LCI results to the selected impact categories, while characterization means converting those results into category indicator outcomes relating to the defined FU [23, 24].

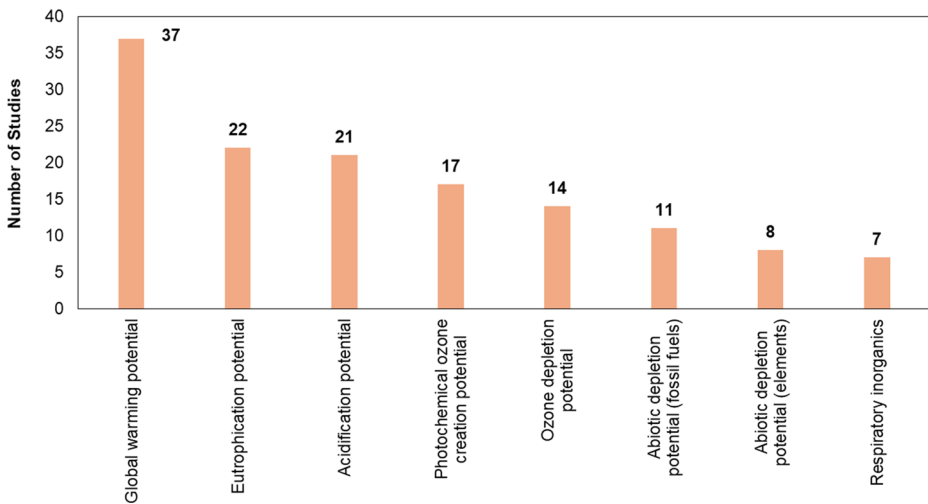
**Life Cycle Impact Assessment Approach** LCIA commonly uses two modeling approaches. The midpoint (problem-oriented modeling) approach evaluates impacts at an intermediate stage along the cause-effect chain and typically includes indicators such as global warming potential (GWP), ozone depletion potential (ODP), acidification potential (AP), and can include more than 15 other indicators [30, 46]. The endpoint (damage-oriented modeling) approach links impacts further down the cause-effect chain to areas of protection such as human health, ecosystem quality, or resource depletion [46]. Among the reviewed studies, 20 did not report the LCIA modeling approach used. However, based on the LCIA methods and the impact categories used, these can be classified as midpoint studies. In addition, 15 studies explicitly reported using a midpoint approach, one reported an endpoint approach, and one used a combined midpoint-endpoint approach.

**Life Cycle Impact Assessment Methods** A wide range of LCIA methods are available, each relying on distinct characterization models that link emissions or resource use to the potential impact categories. In this review, 9 studies (out of 37) did not report the LCIA method used. Among the reported LCIA methods, variations of the CML method were used in 11 instances. Other widely used methods included the IPCC (6 instances), TRACI (4 instances), ReCiPe (3 instances), and IMPACT 2002+ (2 instances). Two instances reported the use of the Environmental Product Declaration (EPD) 2018 [47] and EN 15804:2012+A2:2019 [48] methods. However, these are not typically considered as stand-alone LCIA methods in the same sense as CML or ReCiPe, but rather reporting/standardization frameworks. The mapping of LCIA methods to individual studies is provided in Supplementary Table S6, and their overall distribution is shown in Fig. 7.

A wide range of impact categories is available in LCIA. However, most studies selected a limited set of core categories. The distribution of the frequently used impact categories is shown in Fig. 8. The impact categories and LCIA methods reported in each reviewed



**Fig. 7** LCIA methods and reporting frameworks identified in the reviewed studies (instances,  $N=38$ ). Totals are reported as instances because some studies reported multiple impact assessment methods



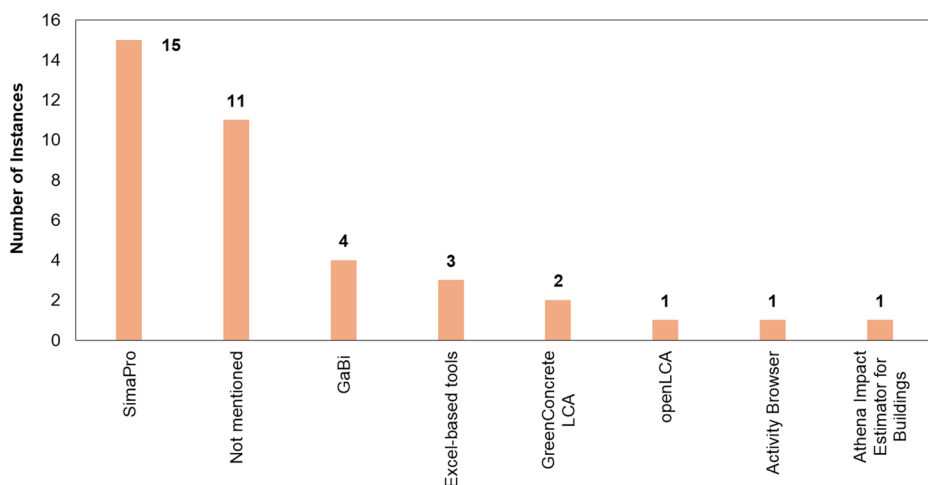
**Fig. 8** Frequently reported midpoint impact categories in the reviewed studies (studies,  $N=37$ )

study are presented in Supplementary Table S6. Since multiple energy-related indicators were identified across the studies, the supplementary table records only whether an energy-related indicator was included in each study.

As shown in Fig. 8, GWP was assessed in all the studies, while other frequently assessed categories include eutrophication potential (EP), AP, photochemical ozone creation potential (POCP), ODP, and abiotic depletion potential (ADP). In addition to standard midpoint categories such as GWP, ODP, and AP, many LCA practitioners also applied standalone indicators to capture energy-related impacts. These include energy demand (ED) [49], cumulative energy demand (CED) [13, 50, 51], energy consumption (EC) [38, 52], and embodied energy (EE) [53, 54]. Another standalone method used was AWARE [13, 55] to assess water scarcity potential. In addition, two studies [45, 56] used mineral resources depletion, while three other studies each adopted waste generation [45], raw material input [55], and eco-cost [57] as standalone indicators.

In line with ISO recommendations, the LCIA phase may also include four optional elements: normalization, weighting, grouping, and data quality analysis [24]. Only a few studies incorporated these additional steps. This review primarily focuses on data quality analysis, which is presented in a dedicated section.

**LCA Software Tools** LCA practitioners use a range of LCA software platforms for inventory modeling and impact assessment (Fig. 9). The most frequently used tool was SimaPro (versions 8.0, 8.4, 9.1, 9.2, and 9.4), reflecting its well-established role as a commercial LCA platform. GaBi was used in 4 instances. A limited number of instances reported the use of customized Excel-based tools (e.g., the modified Gursel 2014 framework) or specialized open-source tools, including openLCA, Activity Browser, and the Athena Impact Estimator for Buildings. The GreenConcrete LCA tool developed by Gursel and Horvath was used in 2 instances. Notably, 11 studies (out of 37) did not mention the software used. An overview of the use of LCA software in the reviewed studies is provided in Supplementary Table S7.



**Fig. 9** Distribution of LCA software tools reported in the reviewed studies (instances,  $N=38$ ). Totals are reported as instances because some studies reported multiple software tools

## Life Cycle Interpretation

According to ISO standards, the interpretation phase of an LCA involves three core elements: (i) identification of significant issues emerging from the LCI analysis and LCIA, (ii) evaluation of the study through completeness, sensitivity, and consistency checks, and (iii) drawing of conclusions, limitations, and recommendations [23, 24].

In the reviewed studies, the interpretation phase was generally used to identify environmental hotspots and to compare the environmental performance of RAC with that of conventional concrete (CC). Cement production was most frequently identified as the dominant contributor to environmental impacts, particularly for GWP, while transportation was also highlighted as an influential process in several studies [26, 41, 50, 56]. For RAC systems, the interpretation of results was commonly linked to RCA replacement level, transport distance, RCA processing method, cement adjustment, and allocation approach [22, 56].

The comparative findings showed that RAC did not consistently perform better than CC across all impact categories [9, 22]. Environmental benefits were more commonly reported when RCA replaced NCA without requiring significant additional cement, when RCA transport distances were relatively short, and when avoided landfill disposal was considered [9, 56]. In contrast, the environmental advantage of RAC was reduced or reversed when additional cement was needed to maintain functional equivalence, when RCA processing was energy-intensive, or when longer transport distances were assumed [22, 41]. Therefore, the environmental performance of RAC should be interpreted as highly scenario- and method-dependent, rather than assumed to be universally beneficial, highlighting the importance of transparent and carefully justified methodological decisions in RAC LCA studies.

## Data Quality Analysis

To enhance the robustness and reliability of LCA outcomes, additional analyses such as sensitivity and uncertainty analyses are often recommended. These analyses help assess the stability of LCA outcomes and the level of confidence with which conclusions can be drawn [25, 30].

Sensitivity analysis examines how changes in input data, assumptions, and methodological choices influence LCA outcomes [24]. In this review, 10 studies conducted sensitivity analysis as presented in Supplementary Table S8. The most frequently examined parameter was transport distance (for RCA or NA), while other parameters included mix design factors (e.g., additional cement demand, RCA proportion, cement type), recycling process configurations (e.g., fixed versus mobile treatment facilities), energy system (electricity grid composition, kiln fuel, renewable sources), different allocation approaches, transport mode/type, and CO<sub>2</sub> uptake through carbonation.

Uncertainty analysis assesses how variability in input data and model assumptions propagates through the LCA model and affects the reliability of outcomes. In this review, only two studies incorporated an uncertainty analysis. Mostert et al. [55] applied it at the building level, while Mistri et al. [58] used a Monte Carlo simulation to assess the influence of input variability in the context of developing India-specific LCA databases. Overall, the limited use of sensitivity and uncertainty analyses suggests that the robustness of RAC LCA conclusions was often only partially evaluated, a point to consider when interpreting the reported environmental benefits and trade-offs.

## Discussion

### Emerging Trends in RAC LCA Practice

The analysis revealed a gradual increase in RAC LCA studies over the past decade, with 68% of studies published between 2021 and 2025. This reflects the increasing recognition of LCA as a tool for assessing the environmental sustainability of recycled materials, in line with global shifts toward CE.

Geographical analysis indicates that China, Italy, and Spain are leading contributors to RAC LCA literature, suggesting that countries with high CDW generation and/or more established regulatory frameworks are shaping current LCA practice. While Europe and East Asia dominate the research landscape, African nations (except Egypt) and most parts of South America, Central and Eastern Europe, and the Middle East are underrepresented. Limited contributions from North America and large parts of the Asia-Pacific region further indicate that the evidence base is largely Euro-Sino-centric. This concentration may constrain methodological diversity and limit the contextual transferability of LCA findings.

The keyword co-occurrence map shows strong thematic clustering around core environmental CE topics. The prevalence of terms such as “life cycle assessment”, “construction and demolition waste”, “recycled aggregate”, “recycled concrete aggregate”, and “sustainability” reflects an emphasis on environmental evaluation of material recovery pathways in recycled concrete production. However, the relative absence of keywords related to social

or economic sustainability, policy integration, data harmonization, or regional adaptation indicates that most studies remain focused on the environmental dimension of sustainability. Future research could therefore benefit from more multidimensional approaches, including stronger integration of socio-economic considerations, context-specific datasets, and policy linkages to support broader sustainability appraisal.

## Methodological Gaps and Reporting Inconsistencies in RAC LCA Practice

This section discusses the key methodological gaps and reporting inconsistencies identified across the reviewed studies. These issues relate to system approach selection, FU definition, system boundary selection, allocation and multifunctionality handling, inventory analysis, impact assessment, software usage, interpretation, and data quality analysis. In the context of CE implementation, these gaps are important because they affect how reliably RAC LCA studies can represent material recovery, functional equivalence, cross-life-cycle burden sharing, waste diversion, avoided use of virgin materials, and EOL recycling. Therefore, the identified methodological limitations should be understood not only as reporting issues, but also as factors that may affect the credibility of CE claims for RAC systems. A summary of these limitations is provided in Supplementary Table S9, while detailed discussions are presented in the following subsections.

### System Approach Selection (Attributional/Consequential)

Although ISO standards emphasize the importance of clearly reporting and justifying the choice between an attributional and a consequential approach, none of the reviewed studies explicitly stated their system approach. Given that the selected approach can substantially influence the interpretation of LCA outcomes [25], clearer reporting would improve methodological clarity and comparability.

The dominance of attributional modeling reflects a prevailing focus on representing existing production processes, rather than explicitly exploring market or behavioral consequences of RCA integration. This may reflect the status of RCA integration into concrete as a developing rather than fully mainstream market, where most LCA frameworks are designed to compare product-level environmental performance under existing conditions. While ALCA provides a robust baseline for assessing and comparing environmental performance, it offers limited insight into the dynamic effects of policy interventions, technology adoption, or market substitution linked to increased RCA use. Consequently, the limited application of CLCA approaches suggests an opportunity for system-level modeling that captures feedback mechanisms such as shifts in aggregate demand, transport logistics, and waste management practices to better inform environmental management and CE decision-making for recycled construction materials.

### Functional Unit (FU) Definition

Among the reported FU instances ( $N = 42$ ), the volumetric FU ( $1 \text{ m}^3$  of concrete) was the most common (50% of instances). However, this FU may not fully capture functional equivalence between RAC and CC [22]. ISO 14040/44 requires that the FU define the product's intended function and quantify it in terms of relevant performance characteristics [23, 24].

In addition, the EN 15804:2012+A2:2019 standard, which is widely applied in EPDs for construction products, classifies volumetric measures such as 1 m<sup>3</sup> of concrete as declared units and recommends their use primarily when the product's function or application is not fully defined [36].

Even where performance differences exist, volumetric comparisons may remain appropriate for applications such as non-structural filling or bulk use, where functional requirements are relatively uniform. However, this approach becomes more problematic for higher-value structural concrete applications, where performance equivalence is critical. In such contexts, functional requirements are typically predefined by design specifications (e.g., a specified strength grade for columns). Therefore, continued reliance on volumetric FU without integrating performance parameters into the FU definition may limit comparability across studies and reduce the decision relevance of comparative LCA conclusions.

This issue is less critical for materials such as steel, where closed-loop recycling and 1:1 substitution are generally feasible without compromising functional performance compared with virgin material. However, in concrete, many studies have shown that a 1:1 volumetric replacement of NCA with RCA often reduces functional performance [42]. If such differences are not considered explicitly, comparative LCAs may be affected by functional inequivalence.

To achieve equivalent performance, RAC mixtures typically require higher cement content or superplasticizers, which alter mix design and influence the overall LCA outcome [44, 45]. Accordingly, defining an FU that integrates performance parameters supports more technically meaningful and environmentally robust comparisons between RAC and CC systems [41].

Performance metrics typically integrated into the FU of RAC LCA studies include compressive strength, workability, and durability [41, 56]. Accordingly, several studies adopted FUs that more closely aligned with ISO principles: strength-based FU (26% of instances), strength–workability FU (7% of instances), and strength–durability FU (7% of instances) (Table 1). In other studies (7% of instances), the environmental impact of 1 m<sup>3</sup> of concrete was normalized to predefined performance parameters. In this review, studies that initially defined a volumetric FU but subsequently ensured functional equivalence through mix design adjustments were categorized accordingly, as their comparative assessments were ultimately performance-consistent. Nevertheless, explicitly defining the intended functional

**Table 3** Proposed tiered functional unit selection approach for RAC LCA research

| FU Type                             | Suitable Applications                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volumetric FU                       | Preliminary screening, generic mix comparison, hotspot analysis, and non-structural applications where functional requirements are relatively uniform |
| Strength-based FU                   | Structural-grade comparison, material substitution, and mix optimization, where strength equivalence is the main requirement                          |
| Strength-durability FU              | Exposure-sensitive structural applications requiring both mechanical and durability equivalence                                                       |
| Strength-durability-service life FU | Decision-grade LCA, infrastructure planning, and whole-life assessment requiring long-term functional equivalence                                     |

performance within the FU definition is recommended by ISO standards to improve clarity, transparency, and interpretability [23, 24].

Among the performance metrics identified, workability is a fresh-state property and does not directly reflect the in-service performance of concrete. In contrast, compressive strength is a key determinant of structural performance, and durability strongly influences life-cycle impacts through differences in maintenance, repair, and service life [41]. Furthermore, EN 15804 standard requires the inclusion of a reference service life within the FU definition to support functional equivalence [36]. Therefore, where sufficient data are available, a strength–durability–service life FU (e.g., 1 m<sup>3</sup> of concrete with equivalent compressive strength, durability, and reference service life) provides a more robust basis for comparing the environmental performance of RAC and CC systems. Based on these considerations, Table 3 presents a tiered FU selection approach for RAC LCA studies, from simple volumetric units to performance- and service-life-integrated FUs.

This tiered approach highlights that FU selection should reflect the intended application, decision context, and availability of data. While volumetric FUs may be sufficient for preliminary or non-structural comparisons, performance-integrated FUs are more appropriate for structural and exposure-sensitive RAC environmental assessments, particularly where CW upcycling is intended.

Mass-based FUs and performance-normalized impact FUs are not emphasized in this tiered framework. Mass-based FUs have limited relevance for RAC comparison because concrete is normally specified and used by volume and performance rather than by mass. Similarly, impact-normalized results, for instance, as impacts normalized by compressive strength, can support interpretation but do not necessarily define an equivalent concrete function unless the FU itself specifies the required strength, durability, exposure condition, and reference service life. Moreover, such normalization may implicitly assume a linear relationship between environmental impact and the selected performance parameter, even though the functional value of concrete may not scale proportionally with that parameter. Therefore, if mass-based or performance-normalized impact FUs are adopted, their selection should be explicitly justified, and comparative results should be interpreted cautiously.

## System Boundary Definition

Among the instances of a reported system boundary ( $N=38$ ), 82% were cradle-to-gate, while 10% were cradle-to-grave and 8% were cradle-to-cradle. In most cases, cradle-to-gate studies considered recycling substitution only at the material input stage and excluded further recycling scenarios. This tendency may reflect ongoing uncertainties regarding the recyclability of RAC at EOL, as material quality has been reported to decline with successive cycles.

The predominance of cradle-to-gate boundaries is justified by several studies based on the following rationale. Some studies assume functional equivalence, suggesting that RAC and CC systems perform similarly during the construction, use, and demolition phases [6, 11]. Others argue that for concretes designed for the same application, evaluating only the production phase is sufficient, as subsequent life-cycle stages are expected to have limited influence on LCA outcomes [56, 59, 60]. In some studies, the later stages were excluded primarily as a system boundary choice [52, 57, 61]. Additionally, certain authors contend that post-gate emissions are minor relative to those from raw material extraction and production

phases [11], while others excluded later phases due to data limitations or unavailability [62]. While such simplifications reduce LCA modeling complexity, they can limit insight into service-life and EOL sustainability trade-offs.

More importantly, the dominance of the cradle-to-gate boundary can influence how RAC is interpreted as a CE strategy. When only production-stage burdens are considered, RAC benefits are often predominantly represented through reduced virgin aggregate extraction and the integration of RCA as an input material. However, this boundary may exclude several mechanisms that are central to circularity, including differences in durability and maintenance requirements during use, carbonation uptake during service and after demolition, EOL crushing and recovery processes, residual waste disposal, and potential avoided landfill burdens. As a result, cradle-to-gate studies may either overstate RAC benefits by ignoring possible service-life or maintenance penalties or understate benefits by excluding carbonation uptake, future material recovery, and avoided disposal impacts. Therefore, CE claims based solely on cradle-to-gate results should be interpreted with caution unless the study's goal is limited to a production-stage comparison and functional equivalence between RAC and CC is clearly justified.

Only a minority of studies attempted full life cycle coverage, including those adopting a cradle-to-grave system boundary. For example, Colangelo et al. [44] omitted the construction, use, and maintenance phases and considered only the demolition stage. Pavlů et al. [6] modeled two use cycles of RAC within an open-loop system, considering demolition and landfill at the end. Alberto López Ruiz et al. [13] omitted the construction, renovation, and use stages, modeling only the waste-disposal stage. Similarly, Aman et al. [63] restricted their assessment to landfill processes and the associated emissions occurring during the EOL stage. Collectively, these examples indicate that even where cradle-to-grave boundaries are adopted, construction and use stages are often excluded, which may limit interpretive depth and the identification of environmental sustainability trade-offs.

RAC can differ from CC in several aspects that may affect these frequently omitted life-cycle stages. RAC typically exhibits higher carbon sequestration potential due to greater porosity and residual cement paste [49, 64], and this potential can vary with RA treatment methods [14, 22]. Durability performance may also vary with exposure conditions [41]. Nevertheless, most LCA studies assume identical service lives under non-aggressive environmental conditions for both RAC and CC [11, 49]. Such assumptions may be reasonable for specific applications where equivalent performance has been demonstrated, but they may also overlook material-specific degradation mechanisms, maintenance requirements, and carbonation behavior. This can lead to simplified representations of the long-term environmental performance of RAC, particularly in structural or exposure-sensitive applications.

There also remains ambiguity in how cradle-to-cradle boundaries are defined and applied. In principle, a cradle-to-cradle boundary should extend beyond the current product life cycle and explicitly represent EOL recovery and the reintegration of recovered materials into a subsequent product system. However, in several reviewed studies, the term cradle-to-cradle was used even when the model primarily represented recycled-material inputs at the production stage. For example, Pu et al. [49] presented their system boundary as cradle-to-cradle; however, the construction, service, and demolition phases were excluded. In practice, the model primarily represented concrete production using RA, including aggregate treatment processes, transport, concrete mixing, and carbonation uptake, rather than a fully modeled cradle-to-cradle concrete life cycle. Therefore, the adopted system boundary appears to align more closely with a cradle-to-gate framework incorporating RCA recycling input.

A similar interpretation appears in Aman et al. [63], which framed the RCA integration modeling scenario as a cradle-to-cradle approach. However, the operational model for this scenario ended at the concrete batching plant due to a lack of data on downstream construction-site transport and CDW incorporation. Thus, although the study conceptually invoked a cradle-to-cradle framework, the modeled system more closely reflects a cradle-to-gate approach incorporating RCA input.

Ding et al. [56] described their study as a closed-loop cradle-to-cradle LCA because CW was modeled as either being landfilled or recycled into RAC. Unlike previous studies that incorporated RCA as a recycled input at the production stage, Ding et al. [56] explicitly included waste-disposal and recycling-related processes at EOL, including landfill burdens and RCA recovery. However, the construction, service, and demolition phases were omitted for modeling simplicity, as they were assumed to be equivalent for CC and RAC under non-aggressive exposure conditions.

These observations indicate that cradle-to-cradle terminology is applied inconsistently across RAC LCA studies. In some studies, it primarily reflects recycled material input at the production stage, while in others it represents partial closed-loop modeling that includes selected EOL or recovery processes. However, these approaches do not always fully represent the broader closed-loop logic typically implied by cradle-to-cradle framing. These inconsistencies highlight the need for clearer and more standardized boundary definitions and reporting, particularly when CE perspectives are invoked.

## Allocation and Multifunctionality Handling

Allocation and multifunctionality handling are among the most challenging methodological aspects in contemporary LCA practice [27, 65]. In principle, closed-loop recycling can often reduce or avoid the need for cross-life-cycle multifunctionality handling, provided that the recycled material displaces an equivalent virgin material and maintains functional equivalence [15]. However, most concrete recycling frameworks are open-loop in nature, with RCA often entering different concrete production systems or being downcycled to lower-grade applications due to reduced material performance [49, 63]. In such cases, transparent treatment of cross-life-cycle burdens and potential credits is required to assign responsibility between the original and subsequent life cycles, as CE systems imply shared responsibility for material recovery. Additional complexity arises from multi-output recycling processes, where several co-products (e.g., RAs, steel, fines, and residual waste) are generated, potentially requiring multi-output allocation.

RAC inherently involves at least two interlinked life cycles and therefore requires a transparent treatment of cross-life-cycle burdens and credits. Despite this, nearly two-thirds of the reviewed studies did not explicitly specify how allocation and multifunctionality were handled. Such omissions can reduce interpretability and transparency, as the selected approach can substantially influence LCA outcomes and comparative conclusions [41]. This transparency gap highlights persistent challenges in ISO-aligned LCA reporting among RAC studies [23, 24].

ISO recommends avoiding allocation where possible, either by dividing unit processes into two or more sub-processes and collecting input–output data for each sub-process, or by expanding the product system boundary (system expansion) to include additional functions related to co-products [24]. Consequently, although system expansion is often discussed in

literature alongside allocation, it is formally an allocation-avoiding approach. When allocation cannot be avoided, it should preferably be based on physical relationships that reflect actual input–output changes. Where such physical relationships cannot be established, allocation may instead be based on alternative criteria, such as economic value [24]. ISO further requires consistent allocation rules across comparable processes, with mass and energy balances maintained between inputs and outputs. These principles also extend to reuse and recycling systems, where clear differentiation between closed-loop and open-loop recycling is needed to ensure methodological consistency and integrity [24]. Whether the methodological issue concerns cross-life-cycle burden and credit sharing or multi-output recycling, the processes subject to allocation, the allocation basis adopted, and the treatment of avoided burdens should be reported transparently to avoid skewed claims about circularity.

To clarify the suitability of commonly used allocation and multifunctionality-handling approaches in RAC LCA, Table 4 summarises their application according to the recycling scenario and type of multifunctionality being addressed.

Among studies that reported allocation or multifunctionality-handling approaches, cut-off and substitution-based avoided-burden approaches were dominant (see Table 2). The cut-off approach considers that the recycled material enters the next life cycle burden-free, with only the recycling-related impacts assigned to the RAC production system [40, 66]. This approach is useful for attributional modeling, particularly when the displaced product is uncertain. However, from a CE perspective, this approach does not explicitly account for shared responsibility between consecutive life cycles or the potential benefits of material substitution, which may limit its ability to represent circular material flows.

Conversely, the substitution-based avoided-burden approach credits the system for avoiding virgin material production and related transport in an alternative or subsequent life

**Table 4** Suitability of common approaches for allocation and cross-life-cycle multifunctionality handling in RAC recycling scenarios

| Approach                          | Suitable RAC Recycling Scenario                                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substitution-based avoided-burden | Suitable for high-quality recycling where RCA functionally displaces NCA or another clearly defined material. Requires justification of the displaced product, substitution ratio, and avoided processes                                                                                              |
| Cut-off                           | Suitable for attributional modeling of open-loop recycling where RCA enters a subsequent product system without carrying previous-life-cycle burdens. Useful when the displaced product or avoided processes are uncertain                                                                            |
| Economic allocation               | Suitable for multi-output recycling or cross-life-cycle burden sharing where allocation is based on the relative market value of outputs. Most appropriate when co-products have established and clearly differentiated markets, but results should be interpreted carefully due to price variability |
| Mass allocation                   | Suitable for multi-output recycling where burdens are distributed among co-products based on measurable physical flows. Applicable when mass provides a reasonable proxy for the relationship between outputs and environmental burdens                                                               |

cycle [9]. This aligns more closely with the CE principles by recognizing the environmental benefits of material substitution. This approach typically reduces the net environmental impacts attributed to recycled systems, thereby favoring recycling-oriented solutions. However, its implementation is sensitive to boundary definitions, substitution ratios, recycling rates, and assumptions about the displaced product. Avoided landfill credits may also be included, but only when recycled CDW would otherwise be landfilled in the absence of recycling into concrete. Where CDW is already diverted to established secondary applications, landfill-avoidance credits may overestimate benefits and inflate the apparent circularity of the RAC system.

Mass and economic allocation approaches provide relatively straightforward burden-sharing mechanisms, distributing impacts according to the physical mass or market value. These approaches are most commonly applied to multi-output recycling processes, although they may also be used for cross-life-cycle multifunctionality handling when physical or economic relationships are explicitly adopted. However, these approaches do not explicitly account for functional differences between RCA and NCA and, when applied without performance-based justification, can lead to conceptually weak or oversimplified allocations [25, 27].

### Life Cycle Inventory (LCI) Analysis

The findings indicate that RAC LCA studies commonly rely on mixed inventory data sources, reflecting both the complexity of RAC systems and the limited availability of standardized datasets for recycling-related processes. The use of primary data in approximately 57% of studies suggests that context-specific data are increasingly recognized as important, particularly for key foreground processes such as waste collection, transport, sorting, crushing, screening, and RCA production.

However, the continued reliance on literature-derived inventory data, used in approximately 70% of studies, raises concerns regarding contextual accuracy and regional representativeness. Several studies were based entirely on literature-derived datasets [22, 41, 52, 53, 61, 63], which may limit their ability to represent specific technologies, operational conditions, regional transport options, the mix of electricity, the specific energy demand and practices of the recycling plant, the composition of the cement, and variability in the quality of RCA. These factors should be considered methodological sensitivity points, as variations can substantially influence comparative LCA outcomes for RAC and CC. As noted by Colangelo et al. [44], relying solely on literature or secondary data can be a methodological limitation of LCA when such datasets are not aligned with the actual system being assessed.

The frequent use of proprietary databases, particularly Ecoinvent, can improve compatibility with major LCA software and provide broad data coverage. However, restricted access to the underlying inventory datasets in proprietary databases may reduce transparency, reproducibility, and cross-study comparability of RAC LCA outcomes. In contrast, the use of regional or public datasets remains comparatively limited, despite their importance for improving geographical representativeness, particularly for electricity generation, transport systems, aggregate production, supply chains, and recycling technologies. Overall, these findings suggest that inventory data quality remains a critical methodological challenge in RAC LCA studies, and that key inventory parameters should be interpreted as sensitivity drivers of LCA rather than only as reporting items.

## Life Cycle Impact Assessment (LCIA)

The findings indicate that RAC LCA studies are strongly dominated by midpoint-based impact assessment, while endpoint and combined midpoint–endpoint approaches remain comparatively underutilized. This pattern is methodologically understandable, as midpoint methods are generally considered more transparent, scientifically robust, and associated with lower uncertainty, whereas endpoint methods provide greater interpretability for decision-makers by translating impacts into damage categories such as human health, ecosystem quality, and resource depletion [67, 68]. However, limited adoption of endpoint modeling may reduce the decision relevance of LCA conclusions, particularly in policy-oriented applications. This is increasingly important because European standards such as EN 15804:2012+A2:2019 recommend a combined midpoint–endpoint perspective to capture both potential environmental pressures and damage to areas of protection [36].

Transparency in LCIA reporting remains a key limitation. Approximately 24% of the studies did not report the LCIA method used. This omission is important because the different LCIA methods assign different characterization factors and impact units to the same impact category. In addition, LCIA methods are developed based on different modeling assumptions, regional scope, midpoint or endpoint structure, and level of uncertainty [69]. When the chosen LCIA method is not clearly reported or justified, these methodological uncertainties may be overlooked, potentially reducing the reliability, transparency, and comparability of the LCA outcome. This is particularly relevant because different characterization models can generate divergent results for the same inventory data, highlighting the influence of LCIA method selection on LCA outcomes [25, 69]. Therefore, cross-study comparison becomes challenging unless the LCIA method is explicitly stated and aligned with the study's goal and scope. Overall, limited disclosure of LCIA methods constrains both the interpretation of individual study results and the development of consistent methodological benchmarks for RAC LCA research.

Among the reported methods, CML (midpoint-oriented) was the most widely used, followed by IPCC, which focuses on climate change using the GWP indicator. Other LCIA methods, including TRACI, ReCiPe, and IMPACT 2002+, were adopted less commonly. TRACI is exclusively midpoint-oriented, while ReCiPe integrates both midpoint and endpoint frameworks. IMPACT 2002+ adopts a hybrid structure that links midpoint indicators to endpoint domains. The prevalence of CML likely reflects its long-standing use, familiarity among LCA practitioners, and broad software compatibility. However, because the CML method does not provide endpoint-level interpretation, relying solely on it may limit broader damage- or policy-oriented interpretation. Accordingly, LCIA method selection should be consistent with the study's goal and scope and explicitly justified to support the credibility and relevance of LCA conclusions [23, 24].

Consequently, the choice of the LCIA method can influence how RAC environmental superiority is interpreted. Different methods and indicator sets vary in category coverage, regional relevance, characterization model, and midpoint or endpoint perspective. As a result, RAC may appear environmentally favorable in categories linked to climate change, resource depletion, or avoided use of virgin aggregate, while trade-offs may emerge in categories affected by additional processing, transport, treatment energy, water use, or toxicity-related impacts. Therefore, conclusions on RAC superiority should be based on a clearly reported and justified LCIA approach and interpreted across relevant impact categories rather than relying on a single indicator.

The climate change impact category, commonly quantified using GWP, was consistently identified as the most frequently assessed environmental impact category. This reflects the well-established contribution of concrete production to greenhouse gas (GHG) emissions and prioritization in environmental assessments [67]. Other frequently reported midpoint categories include EP, AP, POCP, ODP, and ADP. Many studies also emphasized energy use as a major environmental concern, often employing standalone energy consumption indicators. However, reliance on a narrow set of impact categories may limit the ability of RAC LCA studies to capture broader environmental trade-offs associated with RCA integration. Therefore, impact category selection should extend beyond GWP where relevant and be justified according to the environmental mechanisms most affected by the RAC system under study.

The LCA software reporting also has implications for methodological transparency and reproducibility. Although software choice does not by itself determine the validity of an LCA, it influences database access, modeling structure, available LCIA methods, and the reproducibility of inventory and impact assessment procedures. The reviewed studies show a clear dominance of commercial tools, particularly SimaPro, while open-source and concrete-specific tools were used less frequently. This may improve compatibility with established databases and widely used LCIA methods, but it can also limit transparency where proprietary databases or undocumented modeling assumptions are used.

Overall, these findings indicate that LCIA reporting in RAC LCA studies remains uneven, particularly in the explicit declaration and justification of LCIA methods, impact category selection, software tools, and databases. This can limit transparency, comparability, and the decision relevance of LCA conclusions.

## Life Cycle Interpretation

A more comprehensive interpretation phase is needed to move RAC LCA studies beyond descriptive hotspot identification toward clearer decision support. Although many studies have compared RAC and CC environmental performance, the interpretation phase was often limited to reporting which processes contributed most, rather than analyzing the conditions under which RAC becomes environmentally preferable. This is important because the conclusions of the RAC LCA are strongly influenced by factors such as functional equivalence, system boundary selection, allocation and multifunctionality-handling approach, transport distance, RCA processing intensity, and cement content adjustment. Therefore, conclusions on RAC environmental sustainability should not be presented as universal outcomes, but as context-dependent findings shaped by the assumptions adopted in each study. For example, studies should clarify and clearly report whether observed benefits arise from reduced virgin aggregate extraction, avoided waste management, shorter transport distances, material substitution credits, carbonation uptake, or specific modeling assumptions. Such an interpretation would reduce overgeneralization and enhance the usefulness of RAC LCA outcomes for decision-making in circular construction. This would also help distinguish genuine CE benefits from apparent benefits created by narrow system boundaries, unclear allocation approaches, or incomplete EOL modeling.

## Utilization of Data Quality Analysis (Sensitivity and Uncertainty Analyses)

Sensitivity analysis is particularly important in RAC LCA studies because environmental outcomes often depend on parameters that vary across projects and regions. Transportation

was the most frequently examined sensitivity parameter, reflecting its significant influence on the environmental performance of RAC systems, particularly where RCA and NA supply routes differ [56, 70]. RCA proportion was also commonly tested, as it can affect both environmental impacts and mechanical performance, especially when mix design adjustments are required to maintain functional equivalence [40, 55]. Energy-related assumptions, including electricity source, fuel type, and energy mix, were also examined in several studies, highlighting the growing relevance of production energy and decarbonization policy [3, 41]. Importantly, reporting parameters that show limited sensitivity are also useful, as they improve transparency and help future studies identify which variables should be prioritized, validated, or reasonably excluded.

However, uncertainty analysis remains significantly underutilized. This is a significant limitation because RAC LCA studies often combine primary data, literature-derived assumptions, and secondary database values, each of which may introduce variability into LCA modeling. Therefore, wider adoption of uncertainty analysis would improve confidence in LCA conclusions and support more robust comparison and decision-making. Future studies would therefore benefit from more consistent application of sensitivity and uncertainty analyses to key variables, including transport distance, allocation approach, RCA processing energy, cement-content adjustment, substitution ratios, carbonation uptake, and EOL assumptions.

### ISO-Aligned Methodological Recommendations for RAC LCA Practice

Based on the methodological gaps and reporting inconsistencies identified across the reviewed studies, this section presents decision-oriented, ISO-aligned recommendations to improve the robustness, transparency, and comparability of future RAC LCA research. A summary of recommendations for the main LCA components is provided in Supplementary Table S10, while the key recommendations are discussed below.

RAC LCA studies can improve the relevance and transparency of their conclusions by clearly defining the study goal and scope, intended application, decision context, target audience, and whether results are intended to support comparative claims, product development, policy guidance, or preliminary screening. Within this context, studies should explicitly state and justify the use of an attributional or consequential approach. ALCA is appropriate when the study aims to describe or compare the environmental performance of RAC systems under current production and market conditions, whereas CLCA can be adopted when the research question concerns the consequences of decisions likely to affect market conditions, such as large-scale adoption, policy-driven demand changes, shifts in recycling flows, transport logistics, or substitution patterns. Therefore, the choice between ALCA and CLCA should be guided by the study goal, intervention scale, market relevance, and data availability, rather than applying either approach by default.

Volumetric FUs can be strengthened by incorporating relevant performance metrics to improve comparability and ensure that environmental impacts are interpreted on a functionally equivalent basis in line with ISO 14040/44 and EN 15804 guidance. Where RAC and CC differ in strength development, durability performance, or expected service life, these parameters should be incorporated directly into the FU definition rather than discussed only as separate performance indicators.

As many studies adopted cradle-to-gate boundaries, extending system boundaries to capture service-life effects, carbonation uptake, EOL recovery, and subsequent recycling stages may provide a more complete representation of the environmental implications of RCA integration into concrete and material circularity. In addition, transparent and consistently applied definitions of cradle-to-cradle system boundaries are needed to clarify how EOL recycling is modeled, including whether recovered RCA is treated as an output, credited for avoided landfill disposal or virgin aggregate substitution, or reintegrated into a subsequent product system. Consistent with ISO requirements, studies should also clearly define cut-off criteria and report processes excluded from the LCA modeling, as such exclusions can affect the comparability and transparency of RAC LCA outcomes.

Furthermore, the allocation and multifunctionality handling should be explicitly reported and transparently justified, especially when environmental credits are assigned for waste diversion, NA substitution, or avoided landfill. Studies should clearly specify which processes were allocated and how impacts were distributed across life cycles and/or co-products. Future studies should distinguish between scenario analysis, which explores alternative modeling choices such as allocation approach, system boundary, RCA processing route, or EOL pathway, and sensitivity analysis, which examines numerical parameters such as transport distance, RCA replacement ratio, and energy-related assumptions.

The use of primary data for key RAC unit processes can improve the contextual relevance of LCA outcomes, particularly where emissions, transport burdens, and recycling efficiencies vary across regions and facilities. In addition, foreground inventories should clearly report RCA processing steps, transport distances, energy and water inputs, emissions, by-products, and waste outputs to improve inventory completeness and allow meaningful comparison across studies. When secondary databases are used, their regional and technological relevance needs to be clearly justified. Transparent reporting of the software tool, inventory data sources, modeling assumptions, and any modifications made to default datasets or characterization methods is essential for improving reproducibility and comparability across RAC LCA studies. Digital LCA tools and AI-assisted inventory development may also help generate more detailed and harmonized datasets for aggregate recycling systems.

Midpoint LCIA methods may be complemented by endpoint or hybrid approaches where broader interpretation of environmental damage, resource depletion, or pollution-related effects is required. To improve robustness, RAC LCA studies should clearly state the selected LCIA method, justify its relevance to the study context and impact categories considered, and, where appropriate, apply more than one impact assessment method or conduct sensitivity analysis to examine the influence of LCIA method selection. Studies should also consider a comprehensive set of impact categories rather than focusing exclusively on GWP to identify broader environmental sustainability trade-offs associated with RCA integration. The interpretation phase should also move beyond hotspot identification to clarify the conditions under which RAC is environmentally preferable, particularly regarding transport distance, cement-content adjustment, allocation approach, RCA processing intensity, carbonation uptake, and EOL modeling assumptions. Since these factors can alter comparative LCA outcomes between RAC and CC, sensitivity and uncertainty analyses should be treated as essential decision-support tools for testing the robustness of conclusions, identifying influential assumptions, and improving confidence in policy, design, and material-selection decisions.

Finally, while environmental assessment remains central, extending RAC sustainability evaluation to include economic and social dimensions would support a more complete sustainability appraisal of RAC systems. This is especially relevant where the practical adoption of RAC depends not only on environmental sustainability but also on cost competitiveness, supply chain feasibility, and stakeholder acceptance. An integrated assessment approach can therefore provide a stronger basis for decision-making associated with RCA integration.

Overall, these recommendations provide decision-oriented and ISO-aligned practical guidance for LCA practitioners, construction-materials researchers, policymakers, and standardization bodies by supporting more transparent modeling, functionally equivalent RAC comparisons, cautious interpretation of CE claims, and clearer reporting guidance for future RAC LCA studies.

## Conclusions

This study conducted an ISO-informed methodological audit of contemporary life cycle assessment (LCA) practices applied to recycled aggregate concrete (RAC), using a PRISMA-guided systematic review. The review synthesized evidence from 37 peer-reviewed studies published between January 2015 and May 2025 and critically evaluated the prevailing methodological challenges and inconsistencies in LCA practices, including how these limitations may influence the LCA outcomes. By linking the identified limitations to ISO-aligned recommendations, this review provides practical guidance for improving the robustness, comparability, transparency, and credibility of future RAC LCA research and circularity claims. The key conclusions from the review are summarized as follows.

- *Limited transparency in system approach selection affects interpretability:* None of the reviewed studies explicitly stated whether an attributional or consequential LCA approach was adopted, reducing methodological clarity and decision relevance.
- *Over-reliance on volumetric FUs limits functional equivalence:* Frequent use of purely volumetric FUs (1 m<sup>3</sup> of concrete) does not fully reflect performance equivalence between RAC and CC and may reduce the comparability of LCA conclusions.
- *Predominance of cradle-to-gate system boundary limits CE interpretation:* Reliance on cradle-to-gate boundaries restricts the assessment of long-term sustainability trade-offs, including service life effects, carbonation uptake, EOL management, and subsequent recycling stages, and the extent to which RAC systems genuinely support CE outcomes beyond material substitution.
- *Limited reporting and justification of allocation and multifunctionality handling weaken robustness:* Frequent absence of transparent and well-justified treatment of allocation and multifunctionality handling weakens confidence in LCA conclusions, particularly in studies intended to inform policy or CE decisions. Such omissions may also overstate the apparent environmental superiority of RAC when recycling burdens, avoided use of virgin material, landfill diversion, or EOL credits are not consistently accounted for.
- *Inventory data limitations reduce contextual reliability:* Continued reliance on literature-derived and proprietary datasets without sufficient regional or process-specific justification can reduce the representativeness of RAC LCA outcomes.
- *Limited LCIA justification affects decision relevance:* Incomplete reporting or justification of LCIA methods and impact categories can restrict comparability and overlook broader environmental trade-offs associated with RCA integration.

- *Sensitivity and uncertainty analyses remain underused:* Limited testing of key assumptions, such as transport distance, energy mix, cement adjustment, RCA processing, allocation, carbonation, and EOL modeling, can reduce confidence in RAC LCA conclusions.

Collectively, these findings highlight that inconsistent FUs, narrow system boundaries, unclear allocation and multifunctionality-handling approaches, limitations in data quality, limited LCIA justification, and insufficient sensitivity and uncertainty analyses can lead to divergent interpretations of RAC environmental performance. The ISO-aligned recommendations proposed in this review provide a practical basis for improving methodological transparency, reliability, reproducibility, and decision relevance in future RAC LCA research. They also help distinguish genuine CE benefits from outcomes primarily driven by narrow boundaries, unreported allocation assumptions, or incomplete EOL modeling.

This study has some limitations. The review was based on 37 peer-reviewed English-language journal articles identified primarily through Scopus and complemented by backward and forward snowballing. Therefore, some relevant studies that are only indexed in other databases, published in other languages, or in grey literature may not have been captured. Nevertheless, the selected studies provide a strong basis for identifying key methodological trends, reporting gaps, and opportunities to improve ISO-aligned RAC LCA practice.

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## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose.

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